Projections, Gram Matrices, and Statistical Information

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***Abstract*—Several standard quantities in multivariate statistics — the variance inflation factor, partial correlation, the nuisance penalty in the Cramér–Rao lower bound, canonical correlations, and the entries of a Gaussian precision matrix — have the same geometric source. They are ratios of Gram determinants, hence squared distances to subspaces and therefore functions of angle. This note develops that observation from one projection lemma, which identifies orthogonal projection, the Schur complement, and the bordered Gram determinant as three readings of the same construction. The diagonal minor ratio becomes the squared distance from one vector to the span of the others, and the corresponding angle controls the diagonal entry of the inverse.**

In the statistical setting, the Fisher information matrix is the Gram matrix of the score functions in $L^2(P_\vartheta)$, so the Cramér–Rao lower bound factors into a single-parameter bound times $\csc^2\theta_i$, with the efficient score carrying the entire nuisance penalty. For a block of interest parameters, the scalar angle is replaced by the principal angles between score subspaces, the penalty by a product of squared sines, and Hadamard's inequality by Fischer's. For Gaussian vectors, the same geometry yields Hotelling's canonical correlations, Wilks' Λ, entropy factorizations, and the conditional-independence structure encoded by the precision matrix. The results themselves are classical; the contribution here is a single derivation chain, an explicit separation between coordinate-free content and parametrization-dependent artifacts, and numerically stable forms of the resulting diagnostics.

I. Introduction

Several diagnostics in regression, estimation theory, and multivariate analysis have the same algebraic shape: a determinant divided by one of its minors, or a matrix entry divided by the corresponding entry of the inverse. The variance inflation factor, the partial correlation coefficient, the loss of efficiency caused by nuisance parameters, and the conditional variance of one Gaussian coordinate given the rest are all of this form. They are usually introduced separately, in the vocabulary of the field that needs them, and their common origin is left implicit. That origin is elementary: each ratio measures a squared distance from a vector to a subspace, and hence an angle.

The simplest instance already shows the pattern. If v_i is one vector in a family with Gram matrix G, then the ratio $\det G/M_{i,i}$ is the squared length of the residual after projecting v_i onto the span of the other vectors. Equivalently, $\det G/M_{i,i} = \|v_i\|^2 \sin^2\theta_i$, where θ_i is the angle from v_i to that span. This identity immediately explains why the corresponding diagonal entry of G^{-1} blows up when v_i becomes nearly dependent on the rest.

This note develops that identity systematically. Everything rests on one projection lemma: for a family of vectors and one additional vector u, the orthogonal projection of u, the Schur complement in the bordered Gram matrix, and the determinant factorization of that bordered matrix are three readings of the same construction. Once that lemma is in place, the later results follow by choosing what to border: one coordinate gives the diagonal minor ratio, two coordinates give partial correlation, and a full block gives the principal-angle factorization. Because the argument is carried out in an abstract inner-product space, it applies verbatim to Euclidean vectors, to random variables under the covariance inner product, and to score functions in $L^2(P_\vartheta)$.

Two distinctions are emphasized throughout because they are easy to lose. The first is between sequential and simultaneous angles. The determinant factors into a product of angles measured against *predecessors*, which depends on the ordering of the vectors and is therefore not a diagnostic for any particular coordinate; the angle relevant to coordinate i is measured against *all* other vectors and is permutation invariant. A small determinant reports ill-conditioning but does not say which coordinate is responsible. The second distinction is between coordinates and blocks. The angle of a single coordinate against everything else and the principal angles between two blocks answer different questions, so block separation is not a certificate of coordinate-wise identifiability.

In the statistical half of the note, the score functions of a parametric model are treated as vectors in $L^2(P_\vartheta)$ under the inner product $\langle a, b\rangle = \mathrm{E}[ab]$, so that the Fisher information matrix is literally a Gram matrix. The Cramér–Rao bound for a single parameter then splits into the reciprocal of that parameter's own information times a purely geometric penalty $\csc^2\theta_i$, and the residual of the score after projecting out the nuisance directions — the efficient score of semiparametric theory [1], [2] — is exactly the part of the sensitivity that the nuisance parameters cannot imitate. The penalty is a property of the nuisance *subspace* rather than of any parametrization of it, which is the invariance statement that makes the quantity meaningful. For a block of interest parameters, the same argument produces the principal angles [3] between the interest and nuisance score subspaces, the determinant penalty $\prod_k \sin^2\phi_k$, and a worst-case scalar inflation governed by the smallest principal angle. The principal vectors identify the parameter combination that is failing.

Specializing to Gaussian vectors recovers the classical multivariate dictionary. The covariance matrix is the Gram matrix of the coordinates, orthogonal projection coincides with conditional expectation, the diagonal of the precision matrix inverts a conditional variance, and the normalized off-diagonal entries are partial correlations whose vanishing is conditional independence [4]. At block level the principal angles are Hotelling's canonical correlations [5], the determinant factorization is Wilks' Λ [6], [7], Fischer's inequality is the nonnegativity of the mutual information between the blocks, and Hadamard's inequality is the statement that joint entropy is maximized under independence [8]. Each of these is the same identity read under a different inner product.

No claim of novelty is made for the individual results. The contribution is a single derivation chain connecting them, a clean separation between coordinate-free quantities and parametrization-dependent ones, and numerically stable forms of the resulting diagnostics.

The rest of the note is organized as follows. Section II develops the geometry: after fixing the Gram-matrix and angle notation, it establishes the projection lemma, the diagonal minor ratio and its regression reading, the off-diagonal ratio and partial correlation, and the scalar and block forms of the Cramér–Rao bound, closing with the determinant identities of Hadamard and Fischer. Section III specializes the same framework to Gaussian random vectors, where the results become statements about conditional variance, conditional independence, canonical correlation, and entropy. Section IV collects the invariance audit and the computational summary.

II. Minors, angles, and the Cramér–Rao lower bound

A. Setting and notation

Let $v_1, \ldots, v_n$ be vectors in a real inner product space $\mathcal{H}$, and let $G = [G_{i,j}] \in \mathbb{R}^{n\times n}$ be their Gram matrix, defined by $G_{i,j} = \langle v_i, v_j\rangle$. The space $\mathcal{H}$ may be finite-dimensional (such as Euclidean space) or infinite-dimensional (such as the L^2 space of random variables used later for Fisher information).

The Gram matrix notation is used throughout.

Definition 1 (Gram matrix)**.** For vectors $v_1, \ldots, v_r \in \mathcal{H}$, the Gram matrix is the $r \times r$ matrix

$$G^r = \mathrm{Gram}(v_1, \ldots, v_r) = \big[\langle v_i, v_j\rangle\big]_{i,j=1}^{r}$$

In particular, $G = \mathrm{Gram}(v_1, \ldots, v_n)$.

For any $x \in \mathbb{R}^n$,

$$x'Gx = \sum_{i,j} x_i x_j \langle v_i, v_j\rangle = \Big\|\sum_i x_i v_i\Big\|^2 \geq 0$$

Hence $G \succeq 0$ always, and $G \succ 0$ exactly when the v_k are linearly independent. We assume linear independence throughout, so G is invertible. Every principal submatrix of G is itself a Gram matrix of a subfamily and is therefore positive definite. Consequently, all principal minors are strictly positive, so the ratios below are well defined and nonzero.

Write $M_{i,j}$ for the first minor obtained by deleting row i and column j, and $C_{i,j} = (-1)^{i+j} M_{i,j}$ for the corresponding cofactor. The inverse entries satisfy $(G^{-1})_{i,j} = C_{j,i}/\det G = C_{i,j}/\det G$, where the last equality follows because G, and therefore its cofactor matrix, is symmetric.

For $v \neq 0$ and a subspace $S \subseteq \mathcal{H}$, define the angle $\theta(v, S) \in [0, \pi/2]$ by

$$\cos\theta = \frac{\|P_S v\|}{\|v\|},$$

where P_S is the orthogonal projection onto S. Since $v = P_S v + (v - P_S v)$ is an orthogonal decomposition, the Pythagorean theorem gives $\|v\|^2 = \|P_S v\|^2 + \|v - P_S v\|^2$, and therefore

$$\sin\theta(v, S) = \frac{\mathrm{dist}(v, S)}{\|v\|}, \quad \mathrm{dist}(v, S) := \|v - P_S v\|. \tag{1}$$

The same construction applied to a pair of subspaces produces not one angle but a full spectrum of them [3].

Definition 2 (Principal angles)**.** Let $S_a, S_b \subseteq \mathcal{H}$ be subspaces of dimensions p and q, and set $\nu := \min(p, q)$. The principal angles $0 \leq \phi_1 \leq \cdots \leq \phi_\nu \leq \pi/2$ and principal vectors (y_k, z_k) are defined recursively by

$$\cos\phi_k = \max_{\|y\|=\|z\|=1} \big\{\langle y, z\rangle : y \in S_a,\, z \in S_b\big\} = \langle y_k, z_k\rangle \tag{2}$$

subject to $\langle y, y_\ell\rangle = \langle z, z_\ell\rangle = 0$ for $\ell < k$. When $p > q$ we adopt the convention $\phi_k := \pi/2$ for $q < k \leq p$, so that a full list of p angles is always available.

Remark 1 (The two angle definitions agree)**.** For $p = 1$ and $S_a = \mathrm{span}\{v\}$ there is a single principal angle. If $P_{S_b} v \neq 0$, then (2) maximizes $\langle v, z\rangle/\|v\|$ over unit $z \in S_b$, which is attained at $z = P_{S_b} v/\|P_{S_b} v\|$ with value $\|P_{S_b} v\|/\|v\|$; if $P_{S_b} v = 0$ the maximum is 0. In both cases $\phi_1 = \theta(v, S_b)$: the scalar angle (1) is the one-dimensional case of Definition 2, and the two are used interchangeably below. Angles are unsigned throughout—both definitions take a maximum over orientations—so they record the magnitude of an alignment and never its direction.

B. The projection lemma

The next lemma is the core computational step for this section. It links orthogonal projection, the Schur complement, and determinant factorization, with the Schur complement representing residual geometry.

Lemma 1 (Projection, Schur complement, and bordered determinant)**.** *Let $v_1, \ldots, v_m$ be linearly independent with Gram matrix $A \in \mathbb{R}^{m\times m}$. Let $u \in \mathcal{H}$, and define the cross-terms $b_i := \langle v_i, u\rangle$, the energy $\gamma := \|u\|^2$, and the subspace $S := \mathrm{span}\{v_1, \ldots, v_m\}$. Then*

(i) The projection of u onto S is $P_S u = \sum_i c_i v_i$ where the coefficient vector is $c = A^{-1}b$;

(ii) The squared norm of the projection is $\|P_S u\|^2 = b'A^{-1}b$, and the squared distance to the subspace is

$$\mathrm{dist}^2(u, S) = \gamma - b'A^{-1}b := s, \tag{3}$$

which is exactly the Schur complement of A in the bordered Gram matrix $\bar{G} := \begin{bmatrix} A & b \\ b' & \gamma \end{bmatrix}$ of $(v_1, \ldots, v_m, u)$;

(iii) The determinant of the bordered matrix scales by this distance: $\det\bar{G} = s\det A$.

Proof. (i) The projection is uniquely characterized by finding a vector $\hat{u} \in S$ such that the error $u - \hat{u}$ is orthogonal to S.

Orthogonality to S is equivalent to orthogonality to each basis vector v_i. Writing $\hat{u} = \sum_i c_i v_i$, the normal equations become:

$$\begin{aligned} 0 &= \langle u - \hat{u}, v_j \rangle \\ &= b_j - \sum_i c_i \langle v_i, v_j \rangle \\ &= b_j - (Ac)_j, \quad j = 1, \ldots, m, \end{aligned}$$

which simplifies to the matrix equation $Ac = b$. Since $A \succ 0$ is invertible, $c = A^{-1}b$ exists and is uniquely determined.

(ii) Using the relation $Ac = b$ twice, we find the squared norm of the projection:

$$\|\hat{u}\|^2 = c'Ac = c'b = b'A^{-1}b.$$

Since the error vector $u - \hat{u}$ is orthogonal to the projection $\hat{u}$, we apply the Pythagorean theorem to find the squared distance to the subspace:

$$\operatorname{dist}^2(u, S) = \|u - \hat{u}\|^2 = \langle u - \hat{u}, u \rangle = \gamma - c'b = \gamma - b'A^{-1}b.$$

(Expanding algebraically as $\gamma - 2c'b + c'Ac = \gamma - 2b'A^{-1}b + b'A^{-1}b$ gives the same value.) Note that $s > 0$ if and only if $u \notin S$, which matches the linear-independence requirement for the full extended family.

(iii) We can verify the block LDL' factorization (the matrix equivalent of completing the square) by multiplying it out:

$$\bar{G} = \begin{bmatrix} I & 0 \\ b'A^{-1} & 1 \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & s \end{bmatrix} \begin{bmatrix} I & A^{-1}b \\ 0 & 1 \end{bmatrix}$$

where $b'A^{-1}b + s = \gamma$. Since the outer factors are unit triangular, their determinants are exactly 1. By the multiplicativity of determinants, we obtain $\det \bar{G} = s \det A$. $\square$

C. The diagonal minor ratio

Proposition 1 (Diagonal minor ratio)**.** *Let θ_i denote the angle between v_i and the subspace $S_{-i} := \operatorname{span}\{v_k\}_{k \neq i}$ spanned by all other vectors. Then*

$$\frac{\det G}{M_{i,i}} = \operatorname{dist}^2(v_i, S_{-i}) = \|v_i\|^2 \sin^2 \theta_i = G_{i,i} \sin^2 \theta_i, \quad (4)$$

and consequently, the diagonal entries of the inverse Gram matrix isolate this angular dependency:

$$(G^{-1})_{i,i} = \frac{M_{i,i}}{\det G} = \frac{1}{G_{i,i} \sin^2 \theta_i}, \quad \sin^2 \theta_i = \frac{1}{G_{i,i}(G^{-1})_{i,i}} \quad (5)$$

Proof. Let P be the permutation matrix of the transposition (i, n). Then PGP' is the Gram matrix of the family with v_i and v_n interchanged. Because $\det P = \pm 1$, we have $\det(PGP') = (\det P)^2 \det G = \det G$. Deleting the last row and column of PGP' leaves the Gram matrix of $\{v_k\}_{k \neq i}$, whose determinant is $M_{i,i}$; this is unaffected by the internal ordering of the remaining vectors. Hence, we may assume without loss of generality that $i = n$.

Apply Lemma 1 with $A = G^{n-1} = \operatorname{Gram}(v_1, \ldots, v_{n-1})$, $b_i = \langle v_i, v_n \rangle$ for $i = 1, \ldots, n-1$, $\gamma = \|v_n\|^2 = G_{n,n}$, $S = S_{-n}$, $\bar{G} = G$, and $M_{n,n} = \det G^{n-1}$.

(iii) gives $\det G = M_{n,n} s$, and (ii) identifies $s = \operatorname{dist}^2(v_n, S_{-n})$, which by (1) equals $G_{n,n} \sin^2 \theta_n$. This establishes the first equation.

For the second equation, recall from $\det G \cdot G^{-1} = \operatorname{adj}(G)$ that $(G^{-1})_{n,n} = C_{n,n} / \det G$. Since $C_{n,n} = (-1)^{2n} M_{n,n} = M_{n,n}$, we directly get $(G^{-1})_{n,n} = M_{n,n} / \det G = 1/s$. $\square$

The next two remarks provide consistency checks and clarify the statistical meaning of the result.

Remark 2 (Cauchy–Schwarz and the range of the ratio)**.** Applying the Cauchy–Schwarz inequality in $\mathbb{R}^n$ to the dual pair $G^{1/2}e_i$ and $G^{-1/2}e_i$,

$$\begin{aligned} 1 = e_i'e_i &= \langle G^{1/2}e_i, G^{-1/2}e_i \rangle \\ &\leq \|G^{1/2}e_i\| \|G^{-1/2}e_i\| = \sqrt{G_{i,i}(G^{-1})_{i,i}}, \end{aligned} \quad (6)$$

so $G_{i,i}(G^{-1})_{i,i} \geq 1$, confirming that the rearranged form in (5) returns $\sin^2 \theta_i \in (0, 1]$. Equality forces $G^{1/2}e_i \parallel G^{-1/2}e_i$, meaning $Ge_i = \lambda e_i$. In other words, the i-th column of G vanishes off the diagonal, implying $v_i \perp S_{-i}$ and $\theta_i = \pi/2$ (perfect orthogonality).

Remark 3 (Multiple correlation and VIF)**.** Let $g_k := \langle v_k, v_i \rangle$ for $k \neq i$, and let G_{-i} be the Gram matrix of $\{v_k\}_{k \neq i}$. Lemma 1(ii) gives $\|P_{S_{-i}} v_i\|^2 = g'G_{-i}^{-1}g$. The squared multiple correlation—which measures how much variance of v_i is explained by the remaining vectors—is:

$$R_i^2 := \frac{\|P_{S_{-i}} v_i\|^2}{\|v_i\|^2} = \frac{g'G_{-i}^{-1}g}{G_{i,i}} = \cos^2 \theta_i, \quad (7)$$

hence

$$\sin^2 \theta_i = 1 - R_i^2$$

In regression diagnostics, the factor $1/(1 - R_i^2)$ is recognized as the Variance Inflation Factor (VIF). Geometrically, as the vector v_i becomes highly collinear with the other predictors ($R_i^2 \to 1$), the angle θ_i approaches zero. The VIF measures exactly how much the variance of an estimated regression coefficient increases due to this collinearity compared to an orthogonal baseline.

Corollary 1 (Sequential factorization and Hadamard's inequality)**.** *Let $V^{k-1} := \operatorname{span}\{v_1, \ldots, v_{k-1}\}$ (with $V^0 = \{0\}$) and $\psi_k := \theta(v_k, V^{k-1})$. Then the total volume of the parallelotope formed by the vectors is bounded by the product of their lengths:*

$$\det G = \prod_{k=1}^{n} \operatorname{dist}^2(v_k, V^{k-1}) = \prod_{k=1}^{n} G_{k,k} \sin^2 \psi_k \leq \prod_{k=1}^{n} G_{k,k} \quad (8)$$

Proof. We use induction on n with Lemma 1(iii). The Gram matrix of $v_1, \ldots, v_k$ is the bordering of the Gram matrix of $v_1, \ldots, v_{k-1}$ by v_k. Thus, its determinant is the previous determinant multiplied by the squared orthogonal distance $\operatorname{dist}^2(v_k, V^{k-1})$. The base case $n = 1$ is trivially $\det G = \|v_1\|^2$. The inequality follows from $\sin^2 \psi_k \leq 1$, with equality holding throughout if and only if the family is mutually orthogonal. $\square$

There is a fundamental asymmetry between (8) and (4): the angles ψ_k are *sequential* (each vector is evaluated only against its predecessors and therefore depends strongly on ordering), whereas θ_i is defined against *all* other vectors. Only θ_i is permutation invariant, and only it answers how independently coordinate i is determined.

D. Off-diagonal entries: the dual basis and partial correlation

Remark 4 (Dual basis and residuals)**.** Let $r_{i|-i} := v_i - P_{S_{-i}} v_i$ be the residual from Proposition 1, and define the dual family $w_i := \sum_k (G^{-1})_{i,k} v_k$. The inner products of this dual basis with the primal basis yield the Kronecker delta:

$$\begin{aligned}\langle w_i, v_j\rangle &= \sum_k (G^{-1})_{i,k}\langle v_k, v_j\rangle = \sum_k (G^{-1})_{i,k} G_{k,j}\\ &= (G^{-1}G)_{i,j} = \delta_{i,j}, \qquad (9)\\ \langle w_i, w_j\rangle &= \sum_{k,\ell} (G^{-1})_{i,k} G_{k,\ell} (G^{-1})_{\ell,j}\\ &= (G^{-1}GG^{-1})_{i,j} = (G^{-1})_{i,j}, \qquad (10)\end{aligned}$$

showing that G^{-1} is itself the Gram matrix of the dual basis. Moreover, the dual basis vectors are

$$w_i = \frac{r_{i|-i}}{\|r_{i|-i}\|^2}$$

To see this, note that $\langle w_i, v_j\rangle = \delta_{i,j}$ means $w_i \perp v_j$ for every $j \neq i$, hence $w_i \perp S_{-i}$. Also $w_i \in V := \operatorname{span}\{v_1, \ldots, v_n\}$. Since $\dim V - \dim S_{-i} = 1$, the orthogonal complement of S_{-i} inside V is a single line, which contains both w_i and the non-zero residual $r_{i|-i} \neq 0$. Thus $w_i = c r_{i|-i}$. Further, $\langle w_i, v_i\rangle = \langle w_i, r_{i|-i}\rangle = c\|r_{i|-i}\|^2$, which gives the normalization factor $c = 1/\|r_{i|-i}\|^2$.

Taking $j = i$ in (10) recovers $(G^{-1})_{i,i} = 1/\|r_{i|-i}\|^2$, in agreement with Proposition 1. Taking $j \neq i$ and normalizing yields the geometric cosine between residuals:

$$\frac{(G^{-1})_{i,j}}{\sqrt{(G^{-1})_{i,i}(G^{-1})_{j,j}}} = \frac{\langle r_{i|-i}, r_{j|-j}\rangle}{\|r_{i|-i}\|\|r_{j|-j}\|} = \cos\angle(r_{i|-i}, r_{j|-j}) \qquad (11)$$

Proposition 2 (Off-diagonal ratio and partial correlation)**.** *Fix $i \neq j$, let $S_{-i,-j} := \operatorname{span}\{v_k\}_{k\neq i,j}$ (the "control" subspace), and let $r_{i|-i,-j} := v_i - P_{S_{-i,-j}} v_i$ and $r_{j|-i,-j} := v_j - P_{S_{-i,-j}} v_j$ be the residuals after projecting out only the other $n-2$ vectors. Then the partial correlation—the correlation between v_i and v_j after removing the linear effects of all other variables—is:*

$$\begin{aligned}\rho_{i,j|-i,-j} &:= \cos\angle(r_{i|-i,-j}, r_{j|-i,-j})\\ &= \frac{-(G^{-1})_{i,j}}{\sqrt{(G^{-1})_{i,i}(G^{-1})_{j,j}}} \qquad (12)\\ &= \frac{-(-1)^{i+j} M_{i,j}}{\sqrt{M_{i,i}M_{j,j}}}\end{aligned}$$

Proof. It suffices to take $i = n-1$ and $j = n$ by reordering the system. Partition the matrix into blocks accordingly:

$$G = \begin{bmatrix} A & B\\ B' & D\end{bmatrix}, \quad A \in \mathbb{R}^{(n-2)\times(n-2)}$$

where

$$\begin{aligned}B_{k,n-1} &= \langle v_k, v_{n-1}\rangle, \quad B_{k,n} = \langle v_k, v_n\rangle,\\ D_{n-1,n-1} &= \langle v_{n-1}, v_{n-1}\rangle,\ D_{n,n} = \langle v_n, v_n\rangle\\ D_{n-1,n} &= D_{n,n-1} = \langle v_{n-1}, v_n\rangle\end{aligned}$$

and k ranges over the $n-2$ remaining indices.

Block inversion (the 2×2 block version of the LDL$'$ factorization) dictates that the trailing block of G^{-1} is the inverse of the Schur complement:

$$[G^{-1}]_{n-1:n,\, n-1:n} = H^{-1}, \quad H := D - B'A^{-1}B \qquad (13)$$

where H, denoted by $\begin{bmatrix} H_{1,1} & H_{1,2}\\ H_{2,1} & H_{2,2}\end{bmatrix}$, is precisely the Gram matrix of $(r_{n-1|1:n-2}, r_{n|1:n-2})$. Its diagonal entries are the two instances of Lemma 1(ii) with $u = v_{n-1}$ and $u = v_n$; the off-diagonal entry is computed as follows. By Lemma 1(i), $P_{S_{1:n-2}} v_{n-1} = \sum_{k=1}^{n-2} c_k^{(n-1)} v_k$ with $c^{(n-1)} = A^{-1}B_{:,n-1}$. Because the error $r_{n-1|1:n-2}$ is orthogonal to the control subspace $S_{1:n-2}$ while the projection $v_n - r_{n|1:n-2} \in S_{1:n-2}$, we get:

$$\begin{aligned}\langle r_{n-1|1:n-2}, r_{n|1:n-2}\rangle &= \langle r_{n-1|1:n-2}, v_n\rangle\\ &= D_{n-1,n} - (c^{(n-1)})' B_{:,n}\\ &= D_{n-1,n} - B'_{:,n-1} A^{-1} B_{:,n}\\ &= H_{1,2}\end{aligned}$$

Since H is of size 2×2, the adjugate matrix of H is

$$\begin{bmatrix} H_{2,2} & -H_{1,2}\\ -H_{1,2} & H_{1,1}\end{bmatrix}$$

Thus, $\det H$ naturally cancels out in the normalized ratio:

$$\begin{aligned}\frac{-(H^{-1})_{12}}{\sqrt{(H^{-1})_{11}(H^{-1})_{22}}} &= \frac{H_{12}/\det H}{\sqrt{H_{11}H_{22}}/\det H}\\ &= \frac{\langle r_{n-1|1:n-2}, r_{n|1:n-2}\rangle}{\|r_{n-1|1:n-2}\|\|r_{n|1:n-2}\|}\end{aligned}$$

Substituting (13) and rewriting $(G^{-1})_{i,j} = C_{i,j}/\det G$ in cofactor form yields (12). □

The minus sign in (12) is not an arbitrary statistical convention: it comes directly from the off-diagonal sign of the 2×2 adjugate. For the $n = 2$ case of Example 5, comparing (11) with (12) yields the sign flip

$$\cos\angle(r_{1|-1}, r_{2|-2}) = -\cos\angle(v_1, v_2) = -\rho, \qquad (14)$$

since $S_{-1,-2} = \{0\}$ and therefore $r_{1|-1,-2} = v_1$, $r_{2|-1,-2} = v_2$. For general n, the two families of residuals live in different subspaces and their cosines are not simply opposite; the correct unsigned relationship is that the two pairs share the same principal angle (Remark 14). The leading minus sign in the partial-correlation formula compensates for the reversal caused by projecting v_j out of v_i and vice versa.

Example 5 ($n = 2$)**.** For a basic bivariate system $G = \begin{bmatrix} 1 & \rho\\ \rho & 1\end{bmatrix}$, we compute the inverse $G^{-1} = (1-\rho^2)^{-1}\begin{bmatrix} 1 & -\rho\\ -\rho & 1\end{bmatrix}$. Proposition 2 returns $\rho_{1,2} = \rho$, since there is no control subspace to partial out ($S_{-1,-2} = \{0\}$).

Meanwhile, the individual residuals from projecting onto each other are $r_1 = v_1 - \rho v_2$ and $r_2 = v_2 - \rho v_1$. These satisfy $\langle r_1, r_2\rangle = -\rho(1-\rho^2)$ and $\|r_1\|^2 = \|r_2\|^2 = 1-\rho^2$, giving $\cos\angle(r_1, r_2) = -\rho$, which illustrates the sign flip in (14). We also observe $\sin^2\theta_1 = 1-\rho^2$ and $(G^{-1})_{1,1} = 1/(1-\rho^2)$, matching the VIF formulation.

E. The Cramér–Rao lower bound

We now translate this geometry into statistical estimation and quantify how correlated nuisance parameters limit attainable accuracy.

For a parametric model with density $f(x;\vartheta)$, the likelihood function is $L(\vartheta;x) := f(x;\vartheta)$ and the corresponding log-likelihood is $\ell(\vartheta;x) := \log f(x;\vartheta)$. The score function with respect to the i-th parameter is

$$\dot{\ell}_i(x;\vartheta) := \partial_i \log f(x;\vartheta)$$

The collection of score functions forms a family of random variables in $L^2(P_\vartheta)$. For the purposes of this note, the only structure we need is that their Gram matrix under $\langle a,b\rangle = \mathrm{E}[ab]$ is the Fisher information matrix. In the language of information geometry [9], this is the Fisher–Rao metric evaluated on the coordinate score family at ϑ. Throughout this subsection and the next we assume the usual regularity conditions—$f(\cdot;\vartheta) > 0$, differentiation under the integral sign permitted, scores square integrable—and, as the counterpart of the linear-independence assumption of Section II, that $\mathbf{F}(\vartheta) \succ 0$, so that every principal block of $\mathbf{F}$ is invertible.

Definition 3 (Cramér–Rao lower bound)**.** Let $\vartheta \in \mathbb{R}^n$ parameterize a statistical model with Fisher information matrix $\mathbf{F}(\vartheta)$. For any unbiased estimator $\hat{\vartheta}$ of ϑ, the covariance matrix satisfies

$$\operatorname{cov}(\hat{\vartheta}) \succeq \mathbf{F}(\vartheta)^{-1}$$

In particular, the variance of each component obeys

$$\operatorname{var}(\hat{\vartheta}_i) \geq (\mathbf{F}^{-1})_{i,i}$$

and the right-hand side is called the Cramér–Rao lower bound (CRLB) for ϑ_i.

Corollary 2 (Nuisance penalty in the Cramér–Rao lower bound)**.** *Let $\mathbf{F}$ be the Fisher information matrix for a parameter vector $\vartheta \in \mathbb{R}^n$. By definition, $\mathbf{F}_{i,j} = \mathrm{E}\big[\partial_i \log f\, \partial_j \log f\big]$ is the Gram matrix of the score functions $\dot{\ell}_i := \partial_i \log f$ in the Hilbert space $L^2(P_\vartheta)$ under the inner product $\langle a,b\rangle := \mathrm{E}[ab]$. Then any unbiased estimator satisfies $\operatorname{var}(\hat{\vartheta}_i) \geq \operatorname{CRLB}(\vartheta_i)$ with*

$$\begin{aligned}\operatorname{CRLB}(\vartheta_i) &= (\mathbf{F}^{-1})_{i,i} = \frac{M_{i,i}}{\det \mathbf{F}} \\ &= \underbrace{\frac{1}{\mathbf{F}_{i,i}}}_{\substack{\text{single-parameter}\\ \text{bound}}} \cdot \underbrace{\frac{1}{\sin^2\theta_i}}_{\substack{\text{nuisance}\\ \text{penalty}}} = \frac{1}{\mathbf{F}_{i,i}(1-R_i^2)} \end{aligned} \tag{15}$$

Here, θ_i is the angle in $L^2(P_\vartheta)$ between the i-th score and the span S_{-i} of the nuisance scores. The penalty equals unity if and only if the i-th score is orthogonal to all others (parameter orthogonality [2]). It diverges as $\csc^2\theta_i$ in the ill-conditioned regime where $\det \mathbf{F} \to 0$ while $M_{i,i}$ is bounded away from zero.

Proof. Under the regularity conditions above, the score functions lie in $L^2(P_\vartheta)$ with mean zero, $\mathrm{E}[\dot{\ell}_i] = 0$. Consequently, $\mathbf{F}$ serves simultaneously as their covariance matrix and their geometric Gram matrix. The result follows by substituting $v_i = \dot{\ell}_i$ into Proposition 1 and using the multiple correlation equivalence of Remark 3. □

Remark 6 (Efficient score)**.** Let $\dot{\ell}_i^* := \dot{\ell}_i - P_{S_{-i}}\dot{\ell}_i$ be the residual of the i-th score after orthogonally projecting out the nuisance scores. This is known in semiparametric theory [1] as the *efficient score* for the parameter component ϑ_i. By Proposition 1,

$$\operatorname{CRLB}(\vartheta_i) = \frac{1}{\|\dot{\ell}_i^*\|^2} = \frac{1}{\mathrm{E}\big[(\dot{\ell}_i^*)^2\big]}$$

This highlights a key statistical point: the achievable accuracy of estimating ϑ_i is determined not by the total sensitivity of the likelihood to ϑ_i, but *only* by the part of that sensitivity that nuisance parameters cannot mimic. It also shows that the penalty is a property of the *subspace* S_{-i}, not of a particular parametrization of the nuisance block. Any reparametrization that keeps ϑ_i fixed while re-coordinatizing nuisance variables leaves S_{-i}—and therefore the optimal bound—unchanged.

Remark 7 (Information geometry and orthogonal coordinates)**.** From the perspective of information geometry, because $\mathbf{F}(\vartheta)$ acts as the local metric tensor, the angle θ_i measures the geometric orthogonality of the parameter coordinate curves intersecting at ϑ. When $\theta_i = \pi/2$ for all i, the metric tensor is diagonal, meaning the parameters form an orthogonal coordinate system on the statistical manifold. The nuisance penalty $\csc^2\theta_i$ in (15) is the purely geometric cost of using a skewed, non-orthogonal coordinate grid to describe the manifold. Cox and Reid's parameter orthogonality [2] is, geometrically, the search for a local diffeomorphism (reparametrization) that orthogonalizes the tangent basis with respect to the Fisher–Rao metric, thereby eliminating the projection penalty.

Example 8 (Linear Gaussian model / collinearity)**.** Let $y = A\vartheta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0,\sigma^2 I)$ and $A = [\,a_1,\ldots,a_n\,] \in \mathbb{R}^{N\times n}$ having full column rank. Then the log-likelihood is $\log f = -\|y - A\vartheta\|^2/(2\sigma^2) + \text{const}$. The score functions take the form $\dot{\ell}_i = a_i'(y - A\vartheta)/\sigma^2$, leading to:

$$\mathbf{F}_{i,j} = \frac{a_i'\,\mathrm{E}\big[(y-A\vartheta)(y-A\vartheta)'\big]a_j}{\sigma^4} = \frac{a_i'a_j}{\sigma^2}$$

Thus, the Fisher information matrix is the regressor Gram matrix $A'A$ scaled by the noise variance. In this context, the parameter component ϑ_i is associated with the geometric angle in $\mathbb{R}^N$ between the column vector a_i and the span of the remaining columns. Consequently:

$$\operatorname{CRLB}(\vartheta_i) = \frac{\sigma^2}{\|a_i\|^2\sin^2\theta_i} = \frac{\sigma^2}{\|a_i\|^2}\cdot\frac{1}{1-R_i^2} \tag{16}$$

Collinearity appears as a small angle. The regressor energy $\|a_i\|^2$ (representing the signal strength of that variable) may be large, but if the vector lies too close to the hyperplane spanned by the other variables, the parameter estimate remains unreliable.

Remark 9 (Regularization and superresolution)**.** The geometric collapse in Example 8 is exactly why Ridge regression (Tikhonov regularization) adds a penalty term λI to the Gram matrix. Geometrically, this inflates the lengths and opens the angles, bounding the variance at the cost of introducing bias. The same geometry governs closely spaced components in superresolution problems — separating two nearby point sources in optics or in array processing [10] — where $\sin\theta_i \to 0$ as the separation shrinks, so that the error bounds diverge even at fixed signal-to-noise ratio. The block form of this statement is Example 16.

Remark 10 (Diagnostics and computational stability)**.** The Cramér–Rao formulation shows that the determinant $\det \mathbf{F}$ alone—which is maximized in the D-optimality design criterion—cannot pinpoint *which* parameter is poorly determined. By Corollary 1, the determinant aggregates the sequential angles ψ_k into a single overall volume metric. A small parallelotope volume communicates that the system is ill-conditioned, but says nothing about which coordinate is failing.

The true diagnostic quantity is the ratio $M_{i,i}/\det \mathbf{F}$, not either factor in isolation. Numerically, however, computing explicit determinants and minors is unsound: the products underflow or overflow in floating point, and cofactor expansion costs $O(n!)$ operations.

Instead, computationally stable diagnostics compute the Cholesky factorization $\mathbf{F} = LL'$ in $O(n^3)$ operations [11]. One can then obtain the inverse diagonal entries $(\mathbf{F}^{-1})_{i,i} = \|L^{-1}e_i\|^2$ via standard forward/backward triangular solves, and finally read off the geometric angles using $\sin^2\theta_i = 1/(\mathbf{F}_{i,i}(\mathbf{F}^{-1})_{i,i})$.

F. Block parameters and the whitened cross-information

Everything so far has been about a single coordinate. In practice the parameters of interest usually arrive as a group — the position coordinates of one target among several, the regression coefficients of one factor among many, the parameters of one source cluster in Example 16 — and the quantity to be bounded is then a matrix, $\operatorname{cov}(\hat\vartheta_a)$, rather than a single variance. The natural question becomes how far the interest block as a whole is separated from the nuisance block.

One might try to answer it coordinate by coordinate, applying Proposition 1 to each ϑ_i in the interest block in turn. That yields p correct scalar bounds, but they measure each coordinate against *everything else*, including the other interest coordinates, and so report a mixture of two phenomena—coupling to the nuisance parameters, and coupling within the interest block—that a block analysis must keep apart (Remark 12). The separation between two subspaces is not a single number in any case: it is the list of principal angles of Definition 2, introduced in Section II and used here for the first time. This subsection shows that they are exactly the angles carried by the efficient information, so that the scalar factorization (15) becomes a product over principal angles; Propositions 3 and 4 then locate the scalar bound inside the block picture, and Corollary 4 places the resulting determinant identity beside Hadamard's.

1) The efficient information and the whitened cross-information: Partition $\vartheta = (\vartheta_a, \vartheta_b)$ into an interest block of dimension p and a nuisance block of dimension q, block $\mathbf{F}$ conformably, and write $S_a, S_b \subseteq L^2(P_\vartheta)$ for the subspaces spanned by the entries of the score blocks $\dot\ell_a$ and $\dot\ell_b$. Block-inverting $\mathbf{F}$ (the general block form of (13)) gives $(\mathbf{F}^{-1})_{aa} = \mathbf{S}_{a|b}^{-1}$, so the block CRLB for the interest parameters is governed by the Schur complement

$$\operatorname{cov}(\hat\vartheta_a) \succeq \mathbf{S}_{a|b}^{-1}, \qquad \mathbf{S}_{a|b} \coloneqq \mathbf{F}_{aa} - \mathbf{F}_{ab}\mathbf{F}_{bb}^{-1}\mathbf{F}_{ba}, \tag{17}$$

the *efficient information* for ϑ_a; by Lemma 1(ii) it is the Gram matrix of the residuals of $\dot\ell_a$ after projecting out S_b, so (17) is the block form of Remark 6.

Written this way, $\mathbf{S}_{a|b}$ is a difference of matrices and shows no geometry. To expose the angles, put both blocks on a common footing by whitening each with its own information, which turns the coupling term into a cross-Gram matrix of orthonormal bases; principal angles are then computed, rather than found recursively, from its singular values.

Lemma 2 (Singular values of a cross-Gram matrix)**.** *Let $e = (e_1, \dots, e_p)'$ and $f = (f_1, \dots, f_q)'$ collect orthonormal bases of S_a and S_b, and let*

$$K \coloneqq \big[\langle e_i, f_j\rangle\big]_{1\le i\le p,\ 1\le j\le q} \tag{18}$$

be the associated cross-Gram matrix. If $K = YDZ'$ is a singular value decomposition, then $\sigma_k(K) = \cos\phi_k$ for $k = 1, \dots, \nu$, and the principal vectors are the entries of $Y'e$ and $Z'f$.

Proof. For unit vectors $y = \alpha' e \in S_a$ and $z = \beta' f \in S_b$, orthonormality of the bases gives $\|y\| = \|\alpha\|$, $\|z\| = \|\beta\|$ and $\langle y, z\rangle = \alpha' K \beta$. The recursion (2) therefore reads $\cos\phi_k = \max\{\alpha' K\beta \colon \|\alpha\| = \|\beta\| = 1,\ \alpha \perp \alpha_\ell,\ \beta \perp \beta_\ell,\ \ell < k\}$, which is exactly the variational characterization of the kth singular value of K [12, Sec. 7.3], attained at the kth pair of singular vectors. □

Corollary 3 (Angle form of the Gram determinant)**.** *With K as in* (18)*, the eigenvalues of $I - KK'$ are $\sin^2\phi_1, \dots, \sin^2\phi_p$, and consequently*

$$\det\big(I - KK'\big) = \prod_{k=1}^{p}\big(1 - \cos^2\phi_k\big) = \prod_{k=1}^{p}\sin^2\phi_k. \tag{19}$$

Proof. KK' is $p \times p$ with eigenvalues $\sigma_k(K)^2 = \cos^2\phi_k$ for $k \le \nu$ and $p - \nu$ additional zeros, which match $\cos^2\phi_k = 0$ under the convention of Definition 2. □

The next lemma is stated for an arbitrary pair of families, since it is applied twice: to score functions here, and to Gaussian coordinates in Section III-D. Only the Gram structure is used.

Lemma 3 (Whitened cross-Gram matrices)**.** *Let $w_a = (w_{a,1}, \dots, w_{a,p})'$ and $w_b = (w_{b,1}, \dots, w_{b,q})'$ be linearly independent families in a real inner product space, spanning S_a and S_b, with Gram blocks $\Gamma_{aa} = \operatorname{Gram}(w_a)$, $\Gamma_{bb} = \operatorname{Gram}(w_b)$ and $\Gamma_{ab} = \big[\langle w_{a,i}, w_{b,j}\rangle\big]$. Set $u_a \coloneqq \Gamma_{aa}^{-1/2} w_a$ and $u_b \coloneqq \Gamma_{bb}^{-1/2} w_b$. Then the entries of u_a form an orthonormal basis of S_a, those of u_b an orthonormal basis of S_b, and*

$$\Gamma_{aa}^{-1/2}\Gamma_{ab}\Gamma_{bb}^{-1/2} = \big[\langle u_{a,i}, u_{b,j}\rangle\big] \tag{20}$$

is their cross-Gram matrix. Consequently its singular values are $\cos\phi_1, \ldots, \cos\phi_\nu$*, the principal angles being those between* S_a *and* S_b*, and if* YDZ' *is a singular value decomposition of* (20) *the principal vectors are the entries of* $Y'u_a$ *and* $Z'u_b$*.*

Proof. Since $\operatorname{Gram}(w_a) = \Gamma_{aa}$, the Gram matrix of u_a is $\Gamma_{aa}^{-1/2}\Gamma_{aa}\Gamma_{aa}^{-1/2} = I$, so the entries of u_a are orthonormal; they span S_a because $\Gamma_{aa}^{-1/2}$ is invertible. The same argument applies to u_b. Bilinearity of the inner product gives $\big[\langle u_{a,i}, u_{b,j}\rangle\big] = \Gamma_{aa}^{-1/2}\Gamma_{ab}\Gamma_{bb}^{-1/2}$, which is the cross-Gram matrix (18) of two orthonormal bases, so Lemma 2 applies. □

Remark 11 (Any whitening will do)**.** Replacing $\Gamma_{aa}^{-1/2}$ by L_a^{-1} for any factor $\Gamma_{aa} = L_a L_a'$ changes u_a to another orthonormal basis of the same subspace, hence alters the cross-Gram matrix only by orthogonal equivalence and leaves every singular value fixed. Lemma 2 is a statement about S_a and S_b, not about the chosen bases; this is what licenses the Cholesky route of Remark 15.

Applying Lemma 3 to the score families, with $\Gamma = \mathbf{F}$, define the whitened cross-information matrix

$$\mathbf{C} := \mathbf{F}_{aa}^{-1/2}\mathbf{F}_{ab}\mathbf{F}_{bb}^{-1/2} = \mathrm{E}\big[u_a u_b'\big], \tag{21}$$

where $u_a = \mathbf{F}_{aa}^{-1/2}\dot{\ell}_a$ and $u_b = \mathbf{F}_{bb}^{-1/2}\dot{\ell}_b$; its singular values are $\sigma_k(\mathbf{C}) = \rho_k = \cos\phi_k$ for $k = 1, \ldots, \nu$. The Schur complement (17) then factors as

$$\mathbf{S}_{a|b} = \mathbf{F}_{aa}^{1/2}\big(I - \mathbf{C}\mathbf{C}'\big)\mathbf{F}_{aa}^{1/2}, \tag{22}$$

so by Corollary 3 the eigenvalues of $I - \mathbf{C}\mathbf{C}'$ are exactly $\sin^2\phi_1, \ldots, \sin^2\phi_p$. Taking determinants gives the multivariate nuisance-penalty factorization:

$$\det \mathbf{S}_{a|b} = \det \mathbf{F}_{aa}\prod_{k=1}^{p}(1-\rho_k^2) = \det \mathbf{F}_{aa}\prod_{k=1}^{p}\sin^2\phi_k. \tag{23}$$

Each principal angle contributes one geometric penalty term, and small principal angles ($\phi_k \to 0$) create near-nonidentifiable directions in the interest block.

2) Relation to the scalar angle: a variational bracket:

Proposition 3 (Principal angles as stationary values)**.** *For a unit vector* $c \in \mathbb{R}^p$*, let* $\tilde{u}_c := c'u_a \in S_a$*, which has unit norm in* $L^2(P_\vartheta)$*. Then*

$$\cos^2\theta(\tilde{u}_c, S_b) = c'\mathbf{C}\mathbf{C}'c, \tag{24}$$

so the stationary values of $\theta(\cdot, S_b)$ *over the unit sphere of* S_a *are exactly* $\phi_1, \ldots, \phi_p$*, attained at the principal vectors, and*

$$\phi_1 \le \theta(y, S_b) \le \phi_p \qquad \textit{for every } y \in S_a \setminus \{0\}. \tag{25}$$

Proof. Because the entries of u_b are an orthonormal basis of S_b, $P_{S_b}\tilde{u}_c = \sum_k \langle \tilde{u}_c, u_{b,k}\rangle u_{b,k}$ and hence $\|P_{S_b}\tilde{u}_c\|^2 = \|\mathrm{E}[\tilde{u}_c u_b']\|^2 = \|c'\mathbf{C}\|^2 = c'\mathbf{C}\mathbf{C}'c$; dividing by $\|\tilde{u}_c\|^2 = 1$ and using the definition of θ gives (24). The right-hand side is a Rayleigh quotient of $\mathbf{C}\mathbf{C}'$, whose stationary values and extremes are its eigenvalues $\cos^2\phi_k$ by Courant–Fischer. Every $y \in S_a$ is of the form $\tilde{u}_c$ after normalization, which gives (25). □

Remark 12 (What the block and coordinate angles do not share)**.** Taking $a = \{i\}$ and $b = \{1, \ldots, n\} \setminus \{i\}$ gives $p = 1$, $c = 1$, and $\mathbf{C}\mathbf{C}' = \mathbf{F}_{i,-i}\mathbf{F}_{-i,-i}^{-1}\mathbf{F}_{-i,i}/\mathbf{F}_{i,i}$, which by Lemma 1(ii) is exactly R_i^2 of (7); consistently with Remark 1, the single principal angle is θ_i and (23) collapses to Proposition 1.

The converse direction requires care. For $p > 1$, the scalar angles θ_i of the interest coordinates are *not* the principal angles ϕ_k. They are bounded above by ϕ_p, but admit no lower bound in terms of ϕ_1: θ_i is measured against S_{-i}, which contains the other interest scores as well as S_b, whereas (25) concerns angles against S_b alone. Since $S_b \subseteq S_{-i}$ and enlarging a subspace can only decrease the angle to it,

$$\sin\theta_i \le \sin\theta(\dot{\ell}_{a,i}, S_b) \le \sin\phi_p, \tag{26}$$

so a coordinate can be badly determined (θ_i small) even when the two blocks are well separated (ϕ_1 large); the collinearity then lives *inside* the interest block and is invisible to the block geometry. The two diagnostics answer different questions: θ_i asks how well coordinate i is isolated from everything else, ϕ_k asks how well the interest block as a whole is separated from the nuisance block.

Proposition 4 (Block nuisance inflation)**.** *For* $c \neq 0$ *in* $\mathbb{R}^p$*, the bound for the scalar functional* $c'\vartheta_a$ *is inflated by*

$$\mathrm{VIF}(c) := \frac{c'\mathbf{S}_{a|b}^{-1}c}{c'\mathbf{F}_{aa}^{-1}c}, \qquad \frac{1}{\sin^2\phi_p} \le \mathrm{VIF}(c) \le \frac{1}{\sin^2\phi_1}. \tag{27}$$

The upper bound is attained when $\mathbf{F}_{aa}^{-1/2}c$ *aligns with the leading left singular vector of* $\mathbf{C}$*; that is, the worst-case nuisance penalty over all scalar functionals of* ϑ_a *is governed solely by the* smallest *principal angle.*

Proof. Inverting (22) gives $\mathbf{S}_{a|b}^{-1} = \mathbf{F}_{aa}^{-1/2}(I - \mathbf{C}\mathbf{C}')^{-1}\mathbf{F}_{aa}^{-1/2}$. Writing $d := \mathbf{F}_{aa}^{-1/2}c$, the numerator of (27) is $d'(I - \mathbf{C}\mathbf{C}')^{-1}d$ and the denominator is $\|d\|^2$, so $\mathrm{VIF}(c)$ is a Rayleigh quotient of $(I - \mathbf{C}\mathbf{C}')^{-1}$, whose eigenvalues are $1/\sin^2\phi_k$. □

Remark 13 (Invariance: which factor is geometry and which is coordinates)**.** Reparametrize each block separately, $\vartheta_a = T_a\eta_a$ and $\vartheta_b = T_b\eta_b$ with T_a, T_b invertible. The scores become $T_a'\dot{\ell}_a$ and $T_b'\dot{\ell}_b$, so $\mathbf{F}_{aa} \mapsto T_a'\mathbf{F}_{aa}T_a$, $\mathbf{F}_{ab} \mapsto T_a'\mathbf{F}_{ab}T_b$, and $\mathbf{S}_{a|b} \mapsto T_a'\mathbf{S}_{a|b}T_a$: every matrix in (17)–(21) changes. The subspaces S_a and S_b, however, do not, so by Definition 2 the angles ϕ_k are unchanged and $\mathbf{C}$ is altered only by orthogonal equivalence, $\mathbf{C} \mapsto U\mathbf{C}V'$.

Equation (23) therefore separates a coordinate-dependent scale factor $\det \mathbf{F}_{aa}$ from a coordinate-free geometric deficit $\prod_k \sin^2\phi_k$. This is the block form of Remark 6: the penalty is a property of the pair of subspaces, not of the parametrization used to describe them. In fact the ϕ_k are a *complete* invariant: given two pairs of subspaces of the same dimensions (p, q) inside inner product spaces of equal dimension, there is an isometry carrying one pair to the other if and only if their principal angles agree [3]. Nothing about the block coupling is lost by reducing it to these ν numbers.

3) Relation to the determinant identities:

Corollary 4 (Fischer's inequality)**.** *With the convention of Definition 2,*

$$\det \mathbf{F} = \det \mathbf{F}_{aa} \det \mathbf{F}_{bb} \prod_{k=1}^{\nu} \sin^2 \phi_k \leq \det \mathbf{F}_{aa} \det \mathbf{F}_{bb}, \tag{28}$$

with equality if and only if $\mathbf{F}_{ab} = 0$, i.e. $S_a \perp S_b$.

Proof. Lemma 1(iii), in its block form, gives $\det \mathbf{F} = \det \mathbf{F}_{bb} \det \mathbf{S}_{a|b}$; substitute (23), whose padded factors $\sin^2 \phi_k = 1$ for $k > \nu$ contribute nothing. The inequality and its equality case follow from $\sin^2 \phi_k \leq 1$ with equality throughout iff every $\phi_k = \pi/2$. See also [12, Sec. 7.8]. □

Identity (28) places the three determinant statements of this note on a single scale. Corollary 1 bounds the volume by the coordinate energies $\prod_k G_{k,k}$; Corollary 4 bounds it by the block volumes; the exact factorizations identify the deficit in each case as a product of squared sines. Recursively bisecting into 1×1 blocks turns (28) back into (8), so Hadamard is the fully refined instance and Fischer the intermediate one. The refinement is genuinely informative: a system may satisfy Fischer nearly with equality — the two blocks almost orthogonal — while being severely ill-conditioned within a block, exactly the situation flagged in (26).

Remark 14 (Partial correlation is a principal angle, up to sign)**.** Apply Definition 2 to the one-dimensional subspaces spanned by the residuals $r_{i|-i,-j}$ and $r_{j|-i,-j}$ of Proposition 2. There is a single principal angle ϕ, with $\cos \phi = |\rho_{i,j|-i,-j}|$. The absolute value is the point: angles are unsigned (Remark 1), so the sign flip of (14) is invisible at this level — the two residual pairs there have opposite cosines but identical principal angle. Angles report the magnitude of coupling and nothing else; the signed entries of G^{-1} remain necessary wherever direction matters, as in the edge weights of a Gaussian graphical model [4].

Remark 15 (Computing the angles stably)**.** Continuing Remark 10, principal angles should never be obtained from determinants or from explicit matrix square roots. Compute the Cholesky factors $\mathbf{F}_{aa} = L_a L_a'$ and $\mathbf{F}_{bb} = L_b L_b'$, form $\mathbf{C} = L_a^{-1} \mathbf{F}_{ab} L_b^{-\prime}$ by triangular solves — legitimate by Remark 11 — and take its SVD; then $\cos \phi_k = \sigma_k$. This is the Björck–Golub construction [3] and is accurate for angles near $\pi/2$.

Small angles — the diagnostically important regime — are the ill-conditioned case for this route, since $\sigma_k = 1 - \phi_k^2/2 + O(\phi_k^4)$ loses roughly half the available digits, and forming $I - \mathbf{CC}'$ by subtraction compounds the cancellation [13]. Obtain $\sin \phi_k$ instead without subtraction: an LDL' factorization of $\mathbf{F}$ with the nuisance block ordered first yields $\mathbf{S}_{a|b}$ directly as the trailing block, and

$$\sin^2 \phi_k = \lambda_k \big(L_a^{-1} \mathbf{S}_{a|b} L_a^{-\prime} \big). \tag{29}$$

To see (29), substitute (22) and set $M := L_a^{-1} \mathbf{F}_{aa}^{1/2}$; then $MM' = L_a^{-1} \mathbf{F}_{aa} L_a^{-\prime} = I$, so M is orthogonal and $L_a^{-1} \mathbf{S}_{a|b} L_a^{-\prime} = M(I - \mathbf{CC}')M'$ is orthogonally similar to $I - \mathbf{CC}'$, hence has the same eigenvalues. If only the aggregate penalty (23) is wanted, the log-determinants suffice: $\sum_k \log \sin^2 \phi_k = \log \det \mathbf{F} - \log \det \mathbf{F}_{aa} - \log \det \mathbf{F}_{bb}$, each term read off a Cholesky diagonal.

Example 16 (Closely spaced sources)**.** Continuing Remark 9, let ϑ_a collect the parameters of one source cluster and ϑ_b those of another. The score subspaces are spanned by the derivatives of the array manifold at the two parameter values, and ϕ_1 measures how nearly some combination of the first cluster's responses is reproduced by the second. As the separation shrinks, $\phi_1 \to 0$ and Proposition 4 shows the worst-case bound diverging as $\csc^2 \phi_1$ regardless of signal-to-noise ratio, while the leading principal vectors name the specific parameter combination that is failing — information that neither $\det \mathbf{F}$ nor the individual θ_i supplies [10].

III. Application to Gaussian random vectors

The projection-and-Gram framework has a natural application to multivariate normal models. With an appropriate inner product on random variables, algebraic identities for minors and inverses become statistical statements about conditional dependence and optimal prediction.

A. The geometry of conditional expectation

Let $X = [\, X_1, \ldots, X_n \,]'$ be a random vector drawn from a multivariate normal distribution $\mathcal{N}(\mu, \Sigma)$, where we assume the covariance matrix $\Sigma \succ 0$ is strictly positive definite. Without loss of generality, we assume $\mu = 0$, as shifting the mean does not affect variances or covariances.

Consider the Hilbert space $L^2(P)$ of zero-mean, finite-variance random variables, equipped with the covariance inner product:

$$\langle U, V \rangle := \mathrm{E}[UV] = \mathrm{cov}(U, V) \tag{30}$$

This is the same construction used for score functions in Section II-E, with the model distribution in place of P_ϑ. Under this geometry, the covariance matrix Σ is exactly the Gram matrix G of the variables $X_1, \ldots, X_n$. The squared norm of a variable is its variance, $\|X_i\|^2 = \mathrm{var}(X_i) = \Sigma_{i,i}$, and the cosine of the angle between two variables is exactly their Pearson correlation coefficient:

$$\cos \angle(X_i, X_j) = \frac{\langle X_i, X_j \rangle}{\|X_i\| \|X_j\|} = \frac{\mathrm{cov}(X_i, X_j)}{\sqrt{\mathrm{var}(X_i)\,\mathrm{var}(X_j)}} = \rho_{i,j}$$

For Gaussian random vectors, orthogonal projection onto the span of the conditioning variables coincides with *conditional expectation*. The point is not that projection is linear — it always is — but that for a jointly Gaussian vector the conditional expectation is itself linear in the conditioning variables.

If $S_{-i} := \mathrm{span}\{X_k\}_{k \neq i}$ is the subspace spanned by all other variables (denoted collectively as X_{-i}), the projection is

$$P_{S_{-i}} X_i = \mathrm{E}[X_i | X_{-i}]$$

Equivalently, the L^2-best linear predictor is the conditional mean itself.

The same statement holds blockwise, and is needed in Section III-D. Partition $X = (X_a, X_b)$ and apply Lemma 1(i) to each entry of X_a with $S = \operatorname{span}\{X_{b,1}, \ldots, X_{b,q}\}$: the coefficient matrix is $\Sigma_{ab}\Sigma_{bb}^{-1}$, so $\mathrm{E}[X_a \mid X_b] = \Sigma_{ab}\Sigma_{bb}^{-1}X_b$, and the Gram matrix of the residual vector $X_a - \mathrm{E}[X_a \mid X_b]$ is, by Lemma 1(ii) applied entrywise,

$$\Sigma_{a|b} := \operatorname{cov}(X_a \mid X_b) = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}. \tag{31}$$

For jointly Gaussian vectors this residual is independent of X_b, so (31) is the conditional covariance and not merely the covariance of a linear prediction error [7, Ch. 2].

B. The precision matrix and conditional variance

Let $\Theta := \Sigma^{-1}$ be the inverse covariance matrix, also called the *precision matrix*. Applying Proposition 1 in this setting gives a direct statistical interpretation of its diagonal entries.

Corollary 5 (Diagonal of the precision matrix)**.** *The squared distance from X_i to the subspace S_{-i} represents the variance of the residual error, which is exactly the conditional variance of X_i given all other variables:*

$$\operatorname{dist}^2(X_i, S_{-i}) = \mathrm{E}\big[(X_i - \mathrm{E}[X_i|X_{-i}])^2\big] = \operatorname{var}(X_i|X_{-i})$$

Following Proposition 1, the diagonal entries of the precision matrix isolate this conditional variance:

$$\Theta_{i,i} = (\Sigma^{-1})_{i,i} = \frac{1}{\operatorname{var}(X_i|X_{-i})} \tag{32}$$

Using the angular formulation from Proposition 1, we can also express this as

$$\operatorname{var}(X_i|X_{-i}) = \operatorname{var}(X_i)\sin^2\theta_i = \operatorname{var}(X_i)(1 - R_i^2) \tag{33}$$

where R_i^2 is the proportion of variance of X_i explained by all other variables. If X_i is highly correlated with the rest of the network ($\sin\theta_i \to 0$), the conditional variance shrinks toward zero and the precision $\Theta_{i,i}$ diverges.

C. Partial correlation and Gaussian graphical models

We now apply Proposition 2 to interpret the off-diagonal entries of the precision matrix. Let $r_{i|-i,-j}$ and $r_{j|-i,-j}$ be the residuals of X_i and X_j after projecting out the control subspace $S_{-i,-j} := \operatorname{span}\{X_k\}_{k \neq i,j}$. In probability terms, these are the prediction errors

$$r_{i|-i,-j} = X_i - \mathrm{E}[X_i|X_{-i,-j}]$$
$$r_{j|-i,-j} = X_j - \mathrm{E}[X_j|X_{-i,-j}]$$

The correlation between these residuals is the *partial correlation*, denoted $\rho_{i,j|\mathrm{rest}}$. It measures the linear dependence between X_i and X_j that cannot be explained by the other variables.

Corollary 6 (Off-diagonal precision and partial correlation)**.** *The partial correlation between X_i and X_j given all other variables is given by the scaled off-diagonal entries of the precision matrix:*

$$\rho_{i,j|\mathrm{rest}} = \frac{-\Theta_{i,j}}{\sqrt{\Theta_{i,i}\Theta_{j,j}}}$$

This equation underpins Gaussian graphical models (GGMs) [4]: for jointly Gaussian variables, zero correlation implies statistical independence. Therefore, if the partial correlation is zero, the residuals are independent, meaning X_i and X_j are *conditionally independent* given the rest of the network:

$$\Theta_{i,j} = 0 \iff X_i \perp\!\!\!\perp X_j | X_{-i,-j} \tag{34}$$

Geometrically, $\Theta_{i,j} = 0$ means that after projecting out the shared environment $S_{-i,-j}$, the remaining variations of X_i and X_j are orthogonal.

D. Canonical correlations and block dependence

The block geometry of Section II-F transfers verbatim to the Gaussian setting, where it acquires its classical name. Partition $X = (X_a, X_b)$ with $X_a \in \mathbb{R}^p$, $X_b \in \mathbb{R}^q$, and Σ blocked conformably. Since Σ is the Gram matrix of the coordinates under (30), Lemma 3 applies with $\Gamma = \Sigma$: the singular values $\rho_k = \cos\phi_k$ of

$$\mathbf{C} = \Sigma_{aa}^{-1/2}\Sigma_{ab}\Sigma_{bb}^{-1/2} \tag{35}$$

are the *canonical correlations* of Hotelling [5], and the principal vectors are the canonical variates. The scalar case $p = q = 1$ returns the ordinary correlation $\rho_{i,j}$.

Three consequences restate earlier results at block level.

Conditional covariance. Combining (31) with (22), $\Sigma_{a|b} = \Sigma_{aa}^{1/2}(I - \mathbf{C}\mathbf{C}')\Sigma_{aa}^{1/2}$, so the generalized-variance ratio is

$$\frac{\det\Sigma_{a|b}}{\det\Sigma_{aa}} = \prod_{k=1}^{\nu}(1 - \rho_k^2) = \prod_{k=1}^{\nu}\sin^2\phi_k, \tag{36}$$

which is Wilks' Λ [6]; this agrees with (23) because the padded angles contribute unit factors. It is the block form of Corollary 5, and for $p = 1$ it reduces to (33).

Mutual information. Combining Corollary 4 with the Gaussian entropy formula [8, Ch. 8],

$$I(X_a; X_b) = \frac{1}{2}\log\frac{\det\Sigma_{aa}\det\Sigma_{bb}}{\det\Sigma} = -\sum_{k=1}^{\nu}\log\sin\phi_k. \tag{37}$$

Each canonical pair contributes an independent term $-\log\sin\phi_k$, and the contribution diverges as that angle closes. Read in this direction, Fischer's inequality (28) is exactly the statement $I \geq 0$, with equality — independence — if and only if every principal angle is right. This is the block counterpart of the reading of Hadamard's inequality given in Section III-E: Hadamard says joint entropy is maximized by mutual independence of coordinates, Fischer says the same for independence of the two blocks, and $\prod_k \sin^2\phi_k$ measures the shortfall.

Conditional independence of blocks. Let $\Theta = \Sigma^{-1}$ and consider the residual subspaces obtained by projecting out X_k, $k \notin \{a, b\}$. Their cross-Gram structure is that of the conditional covariance H of (X_a, X_b) given the rest, and by block inversion (13), $H = \big(\Theta_{[ab],[ab]}\big)^{-1}$. Canonical correlations are invariant under this inversion: whitening so that H has identity diagonal blocks and cross block $R = \operatorname{diag}(\rho_k)$, the inverse has blocks $(I - R^2)^{-1}$ on the diagonal and $-R(I - R^2)^{-1}$ off it, whose whitening returns $-R$. Hence the *partial* canonical correlations are the singular values of

$$\Theta_{aa}^{-1/2}\Theta_{ab}\Theta_{bb}^{-1/2}, \tag{38}$$

and the sign reversal seen there is the block form of the one recorded in (14). In particular $\Theta_{ab} = 0$ if and only if all partial principal angles equal $\pi/2$, if and only if $X_a \perp\!\!\!\perp X_b$ given the remaining variables — the block generalization of (34), and the basis for block-structured penalties in graphical model selection.

E. Information geometry and entropy

Finally, Corollary 1 can be interpreted in information-theoretic terms. The differential entropy of an n-dimensional Gaussian vector is

$$h(X) = \frac{1}{2}\log\det(2\pi e\Sigma) = \frac{n}{2}\log(2\pi e) + \frac{1}{2}\log\det\Sigma$$

Because $\det\Sigma$ factors sequentially into the squared distances (conditional variances), we have

$$\det\Sigma = \prod_{k=1}^{n}\operatorname{var}(X_k|X_1,\ldots,X_{k-1})$$

Taking the logarithm transforms this geometric volume factorization directly into the chain rule of conditional entropy:

$$h(X_1,\ldots,X_n) = \sum_{k=1}^{n} h(X_k|X_1,\ldots,X_{k-1})$$

Hadamard's inequality ($\det\Sigma \le \prod_{k=1}^{n}\Sigma_{k,k}$) geometrically states that the volume of a parallelotope is maximized when its sides are orthogonal. Information-theoretically, this is the maximum-entropy property of the independent Gaussian: for a given set of marginal variances, the joint entropy is maximized when the variables are mutually independent [8, Ch. 8].

IV. Conclusion

Three threads run through this note, and it is worth separating them explicitly.

One derivation chain. Lemma 1 is the only substantive computation. Every subsequent identity is obtained by choosing what to border: a single coordinate against its predecessors gives Hadamard (8); a single coordinate against all others gives the minor ratio (4), hence the VIF, the CRLB nuisance penalty (15), and the diagonal of a Gaussian precision matrix (32); two coordinates against the remaining $n-2$ give partial correlation (12) and the off-diagonal precision entries; and an entire block against its complement gives the principal-angle factorization (23), hence Fischer (28), Wilks' Λ (36), and the Gaussian mutual information (37).

What survives reparametrization. The principal angles ϕ_k, and the scalar angles θ_i under block-diagonal reparametrizations that fix the coordinate in question, are properties of subspaces and are therefore coordinate-free; they are moreover a complete invariant of the pair of subspaces (Remark 13). The scale factors $\det\mathbf{F}_{aa}$, $\mathbf{F}_{i,i}$, and $\|a_i\|^2$ are not: they change under any rescaling of the parameters, which is why a bare determinant or a bare diagonal information entry is not a diagnostic. Each factorization in this note has been written so that the two kinds of factor appear separately.

What to compute. No diagnostic here should be evaluated from its definitional formula. Cholesky factorization and triangular solves give $(\mathbf{F}^{-1})_{i,i}$ and hence θ_i (Remark 10); the Björck–Golub SVD gives the principal angles near $\pi/2$, and the subtraction-free form (29) gives them in the small-angle regime where the SVD route loses half its digits (Remark 15).

The framework is limited to second-order local geometry: the Fisher information is a Gram matrix because it defines the relevant quadratic form at a fixed parameter value. Bounds that are not of this type—Barankin-style bounds, or effects driven by global features of the Fisher–Rao geometry rather than a single local metric tensor [9] — are not captured by one Gram matrix, and the extent to which the angle picture survives in those settings is not addressed here.

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